Dynamics of Financial Correlation Networks in S&P 500 stocks: Topology, Stress and Portfolio Analysis

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ABSTRACT

In this project, I study the dynamics of correlation-based financial networks in the S&P 500 from 2000 to 2026, spanning two major market crashes: the 2008 global financial crisis and the COVID-19 pandemic of 2020. Using a rolling window of 750 trading days with a 5 day shift, I construct and compare the minimum spanning tree (MST) against its calibrated one-factor CAPM counterpart to examine the time varying topology of equity co-movements across 335 S&P 500 stocks. The simulated one-factor CAPM model serves as a topological null hypothesis, enabling structure driven by the market factor to be separated from genuine sectoral organisation. The empirical MSTs exhibit clear hierarchical multi-sector clustering that evolves distinctly during market stress, while the one-factor universe produces a star-like topology regardless of market conditions. To assess economic implications of the network structure, the analysis is extended to portfolio optimisation, to show that minimum-risk assets are consistently located on the outer leaves of the empirical MST.

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I. Introduction

Financial markets are evolving complex systems in which large numbers of heterogeneous agents interact, giving rise to emergent collective behaviour and feedback loops that cannot be easily inferred from individual components alone. A natural consequence of this is that stock returns are not independent, they exhibit persistent pairwise correlations driven by shared macroeconomic exposures, sectoral relationships, shifts in market influence and supply chain linkages. Understanding the structure of these correlations, and how it evolves through time, is central to both the science of financial markets and practical problems of risk management and portfolio allocation.

Network Science has emerged as an important tool to study various complex phenomena, including the correlation structure in financial systems. Early work by Mantegna [1, 2] showed that a correlation matrix of stock returns encodes a hierarchical taxonomy of equities similar to the sectoral organisation of the economy. He achieved this by defining a correlation based distance metric to build a minimum spanning tree (MST) which revealed these hierarchical relationship. Onnela et al. [3–5] extended this static picture to a dynamic rolling-window framework, introducing scalar summaries of the MST like the normalised tree length and mean occupation layer that contract sharply during market crisis and recover thereafter. He also showed that the rolling MST exhibits a scale-free distribution with a power law. Bonanno et al. [6, 7] further demonstrated that returns simulated from a one-factor CAPM model, despite explaining a large fraction of individual pairwise correlations, produces a qualitatively different star-like topology with none of the multi-cluster hierarchy found in empirical data. Hence, the one-factor CAPM model acts as an ideal topological null hypothesis. Tomaso Aste [8, 9] generalised the MST by introducing the Planar Maximally Filtered Graph (PMFG) which retains $3(N-2)$ edges subject to a planarity constraint. He also showed that stocks located at the periphery of a PMFG or MST tend to form better-performing and more diversified portfolios than those at the centre. More recently, Chakraborti [10, 11] and Kukreti [12] synthesis these approaches, and use entropy based measures to continuously monitor the network state and provide early warning signals of market stress.

A fundamental question in financial network analysis is the extent to which the observed correlation structure reflects genuine multi-sector economic organisation rather than being attributed to a common market factor, proxied here by market index returns. In this project, I address this by investigating the dynamics of correlation-based financial networks in the S&P 500 from January 2000 to April 2026, a period that encompasses two major market crises. I construct rolling MSTs (750 day window, 5 day shift) from the returns of 335 continuously traded stocks, and compare their topological evolution against one-factor CAPM counterparts. Wherever the two universes agree, the signal is attributable to the market factor; wherever they diverge, the real market encodes structure that a single factor cannot explain. Topological measures including the degree distribution, survival ratio, mean occupation layer, and weighted portfolio layer are used to characterise the real market and compare it against the one-factor baseline.

II. Data & Methodology

A. Data Description

The adjusted closing price stock data used in this study were collected from the Yahoo finance database for the United States of America (USA) S&P 500 index from 3rd January 2000 to 1st April 2026. The raw dataset was systematically cleaned to maintain consistency and ensure reliability. Only stocks that were continuously traded throughout the entire

sample period were retained in the panel. Further, any stock with missing values or exhibiting highly non-stationary behaviour (price returns exceeding ±60%) was excluded from the dataset. This filtering procedure resulted in a final panel of 335 stocks spanning all 11 sectors of the Global Industry Classification Standard (GICS): Information Technology, Health Care, Financials, Consumer Discretionary, Communication Services, Industrials, Consumer Staples, Energy, Utilities, Real Estate, and Materials. The S&P 500 index adjusted closing price was also obtained from the same database over the identical sample period.

B. Return Series, Cross-Correlation and Distance Matrices

The adjusted closing price for each stock and the index is transformed into a time series of simple logarithmic returns to ensure weak-stationarity and comparability across assets, defined as

$$r_i(\tau) = \ln P_i(\tau) - \ln P_i(\tau - 1) = \ln\left(\frac{P_i(\tau)}{P_i(\tau - 1)}\right), \tag{1}$$

where $P_i(\tau)$ denotes the adjusted closing price of stock (or index) i on trading day τ. To capture the time varying nature of market correlations, the analysis is conducted over a sequence of $M = 1170$ overlapping windows, each spanning $T = 750$ trading days and shifted by $\delta T = 5$ trading days at each iteration. The equal-time Pearson correlation coefficient between the return series of stocks i and j is calculated as

$$\rho_{ij}^t = \frac{\langle r_i^t r_j^t \rangle - \langle r_i^t \rangle \langle r_j^t \rangle}{\sigma_i^t \, \sigma_j^t}, \tag{2}$$

where $\langle \cdot \rangle$ denotes a time average over the T trading days in window t, σ_i^t denotes the standard deviation of the return series of stock i within window t, and $\rho_{ij}^t \in [-1, 1]$. Further, each correlation matrix was transformed into the "ultrametric" distance matrix [1] using the distance measure given by

$$d_{ij}^t = \sqrt{2(1 - \rho_{ij}^t)}, \tag{3}$$

where $d_{ij}^t \in [0, 2]$ for all pairs (i, j) and all windows t. The resultant sequence of correlation and distance matrices forms the input to the network construction procedures described in the following subsection.

C. One-Factor Model

To assess whether the topological features extracted from the real market data reflect genuine multi-sector economic structure or merely the aggregate market factor, a synthetic one-factor CAPM universe is constructed as a calibrated baseline [6]. For each stock i, the following OLS regression is estimated over the full sample:

$$r_i(t) = \alpha_i + \beta_i r_M(t) + \varepsilon_i(t), \tag{4}$$

where $r_M(t)$ denotes the log-return of the S&P 500 index on day t, α_i is the stock-specific intercept, β_i is the market beta, and $\varepsilon_i(t)$ is the residual. The residual variance $\hat{\sigma}^2_{\varepsilon,i}$ is obtained from the regression. Synthetic return series are then generated as

$$r_i(t) = \hat{\alpha}_i + \hat{\beta}_i r_M(t) + \varepsilon_i(t), \qquad \varepsilon_i(t) \sim \mathcal{N}(0, \hat{\sigma}^2_{\varepsilon,i}), \tag{5}$$

where $\varepsilon_i(t)$ is drawn independently across stocks and time steps. This preserves each stock's sensitivity to the market factor and its idiosyncratic noise variance, while suppressing all

cross-sectional dependencies between stocks. The synthetic data is then processed through an identical network construction pipeline as the real data, enabling a direct topological comparison.

D. Construction of the Minimum Spanning Tree (MST)

The minimum spanning tree is constructed from the distance matrix d_{ij}^t, which encodes the interaction strength between pairs of stocks in a given time window, such that all N nodes are connected with exactly $N-1$ edges under the constraint that the total distance is minimized. Kruksal's algorithm is used to obtain the MST from the distance matrix. Two key advantages of the MST are that it constructs a network structure without requiring any arbitrary threshold, and it possesses an inherent hierarchical clustering property. As a parsimonious representation of correlation structure, the MST reduces the number of pairwise distances from $N(N-1)/2$ to $N-1$, retaining only the most significant pairwise relationships.

E. Topological Measures

To characterise the time-varying topology of the MST, three scalar measures are employed; single step survival ratio, mean occupation layer and weighted portfolio layer [4].
The *single-step survival ratio* measures the fraction of MST edges that persist between two consecutive windows,

$$\sigma(t) = \frac{1}{N-1}|E(t) \cap E(t-1)|, \tag{6}$$

where $E(t)$ denotes the edge set of the MST at window t and $|\cdot|$ denotes set cardinality. A drop in $\sigma(t)$ signals rapid topological reorganisation of the tree, as observed during periods of market stress.
At each window t, the *central vertex* v_c^t is identified as the node with the highest betweenness centrality, defined as

$$v_c^t =_v \sum_{s \neq v \neq r} \frac{\eta_{sr}(v)}{\eta_{sr}}, \tag{7}$$

where η_{sr} is the total number of shortest paths from node s to node r, and $\eta_{sr}(v)$ is the number of those paths that pass through v. The central vertex serves as the root of the tree against which all node depths are measured, and is recomputed at every window to reflect the evolving market structure.
The *mean occupation layer* (MOL) is defined as

$$l(t, v_c) = \frac{1}{N} \sum_{i=1}^{N} \mathrm{lev}(v_i^t), \tag{8}$$

where $\mathrm{lev}(v_i^t)$ denotes the graph-theoretic depth of node v_i relative to v_c^t, measured in integer hops. The MOL captures the spread of nodes across the tree: low values indicate a topologically contracted tree concentrated around the hub, while high values reflect a more dispersed, hierarchically rich structure.
To assess the economic implications of the network structure, the *weighted portfolio layer* is defined as

$$l_P(t, \theta) = \sum_{i \in P(t,\theta)} w_i \, \mathrm{lev}(v_i^t), \tag{9}$$

where $P(t, \theta)$ denotes the Markowitz portfolio (A1) at window t parameterised by $\theta \in [0, 1]$,

and $w_i \geq 0$ are the portfolio weights satisfying $\sum_i w_i = 1$. For this study, $\theta = 0$ which yields the minimum-risk portfolio. The weighted portfolio layer measures the average radial position of portfolio assets within the tree, and is compared against $l(t, v_c)$ to assess whether minimum-risk stocks are systematically located on the outer leaves of the MST.

III. Results & Discussion

The results below present findings from real market data and its one factor counterparts, along with insights from their comparison.

A. Correlation Dynamics

Figure 1 shows the average correlation of S&P 500 stock log returns for each time window over the full sample period, together with the corresponding one-factor model universe.

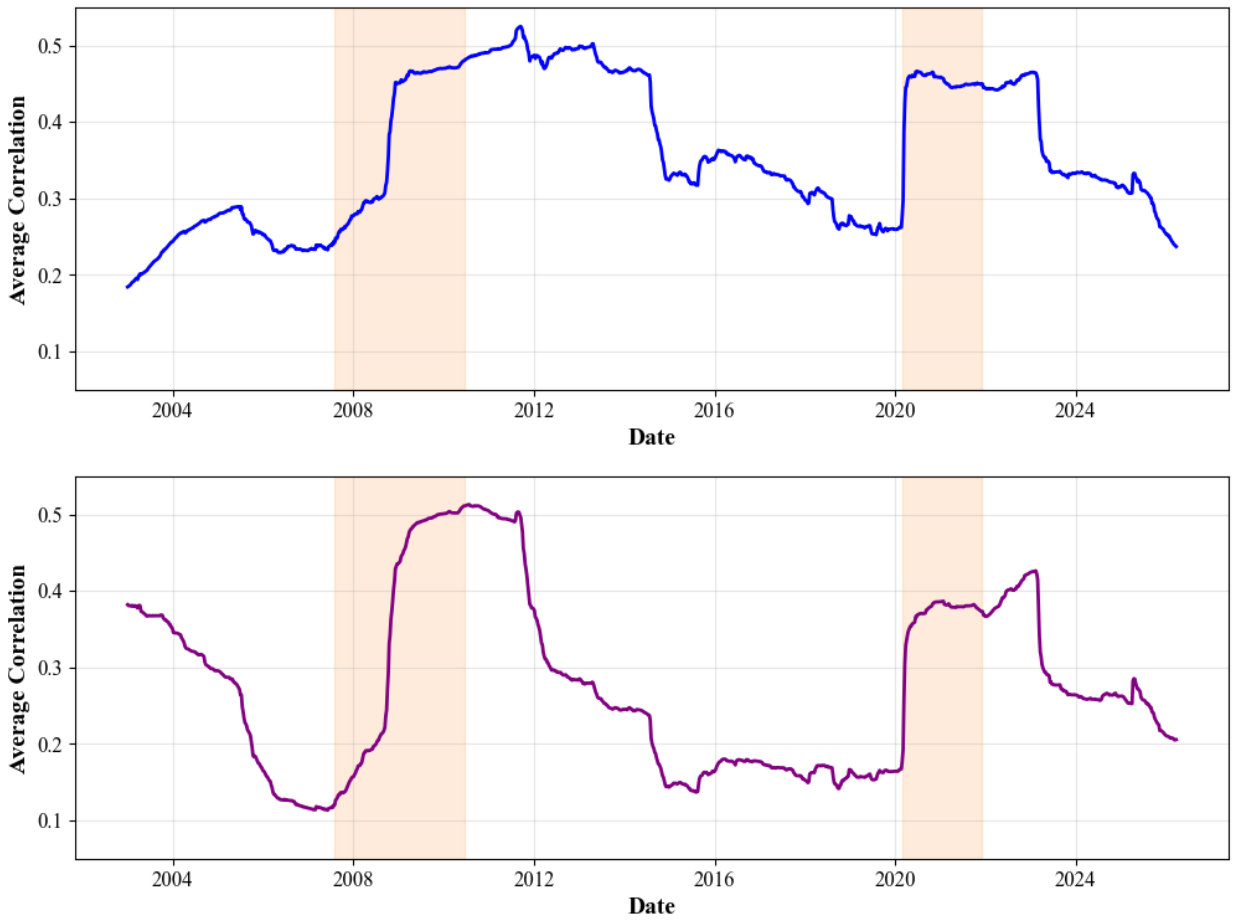

FIG. 1. Comparison of average correlation dynamics. The top figure shows the average correlation of S&P 500 stocks over the sample period, while the bottom figure shows the corresponding one-factor model counterparts. The shaded orange regions correspond to 2 periods of market crises, the 2008 global financial crisis and COVID-19 pandemic of 2020.

Both plots show elevated average correlations during periods of market stress, such as the 2008 global financial crisis and the COVID-19 pandemic in 2020, confirming that the one-factor model provides a valid baseline. During normal periods, empirical correlations deviate from the one-factor model due to idiosyncratic shocks and lower market wide movement.

Figure A1 compares correlation matrices between the empirical S&P 500 stocks and their corresponding one-factor model counterparts across different market conditions. The empirical correlation matrices show clear sectoral block structures, reflecting strong intra-sector correlations, whereas these structures are largely absent in the one-factor model.

B. Network Dynamics

Figure 2 contrasts the MST network topology of the empirical S&P 500 market with the one-factor CAPM universe across three distinct market periods, normal period of December 2005, and crises periods of June 2010 and November 2021.

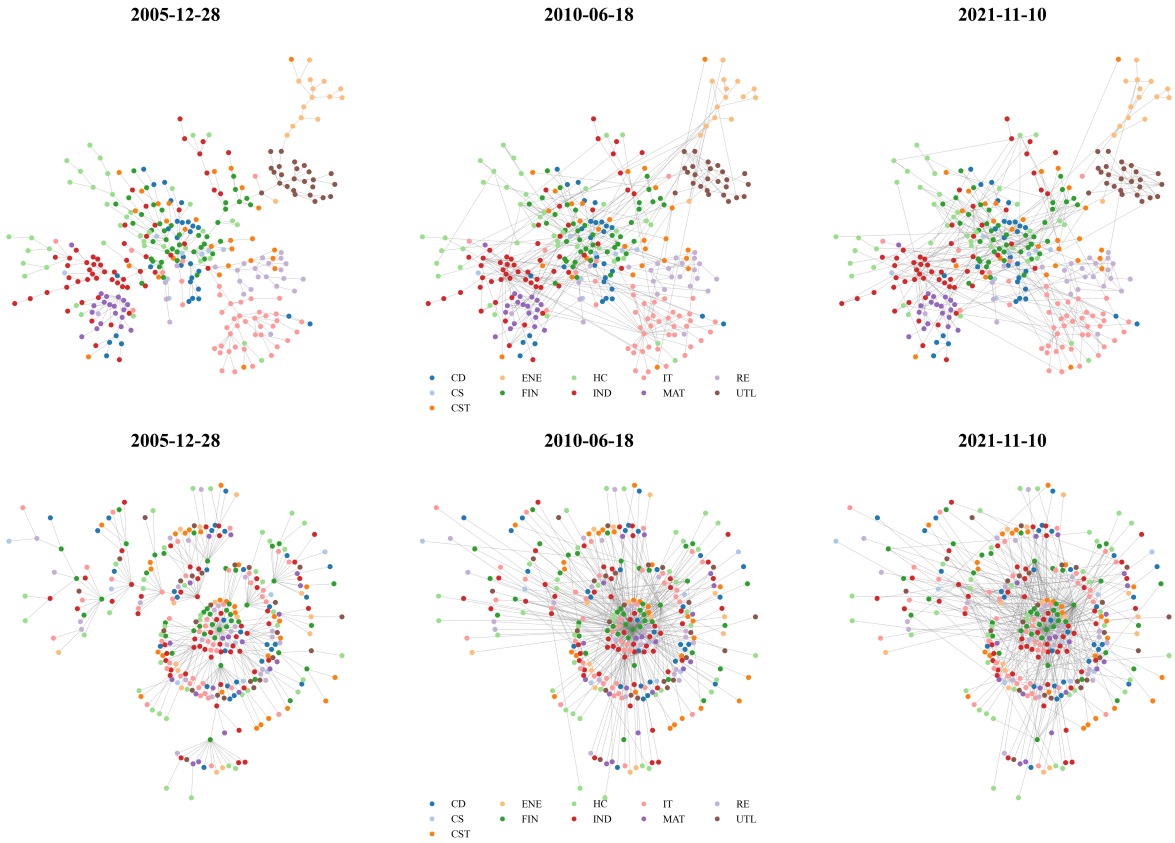

FIG. 2. Comparison of MSTs across different periods. The top panel portrays the network dynamics of S&P 500 market on 28 December 2005 (normal), 18 June 2010 (crash), and 10 November 2021 (crash). The bottom panel shows the corresponding network dynamics generated by the one factor model. Nodes are coloured by GICS sector: Consumer Discretionary (CD, navy), Consumer Staples (CS, light blue), Consumer Staples (CST, orange), Energy (ENE, light orange), Financials (FIN, dark green), Health Care (HC, green), Industrials (IND, red), Information Technology (IT, pink), Materials (MAT, purple), Real Estate (RE, lavender), and Utilities (UTL, brown).

During the normal period, the empirical MSTs exhibits clear sectoral sub-trees, with stocks forming identifiable peripheral clusters. In contrast during the crisis windows, the network becomes denser and edges proliferate across sectoral boundaries, reflecting the elevated market-wide correlations that persist in the aftermath of systemic shocks. Critically, even during these stress periods, the market retains residual sector structure that evolves

meaningfully across market conditions. The one factor model consistently produces a star-like structure across all three windows: a dense, centrally-connected core with nodes radiating outward in a near-symmetric pattern and no discernible sectoral clustering. This is expected, as all synthetic returns share a single common factor, resulting in a near rank-one correlation structure that appears as a hub-and-spoke network regardless of market conditions.

C. Degree Distribution

Figure 3 shows the degree distributions of the MSTs for the real S&P 500 market (left) and the one-factor CAPM universe (right) on a log-log scale, during normal and crisis period windows.

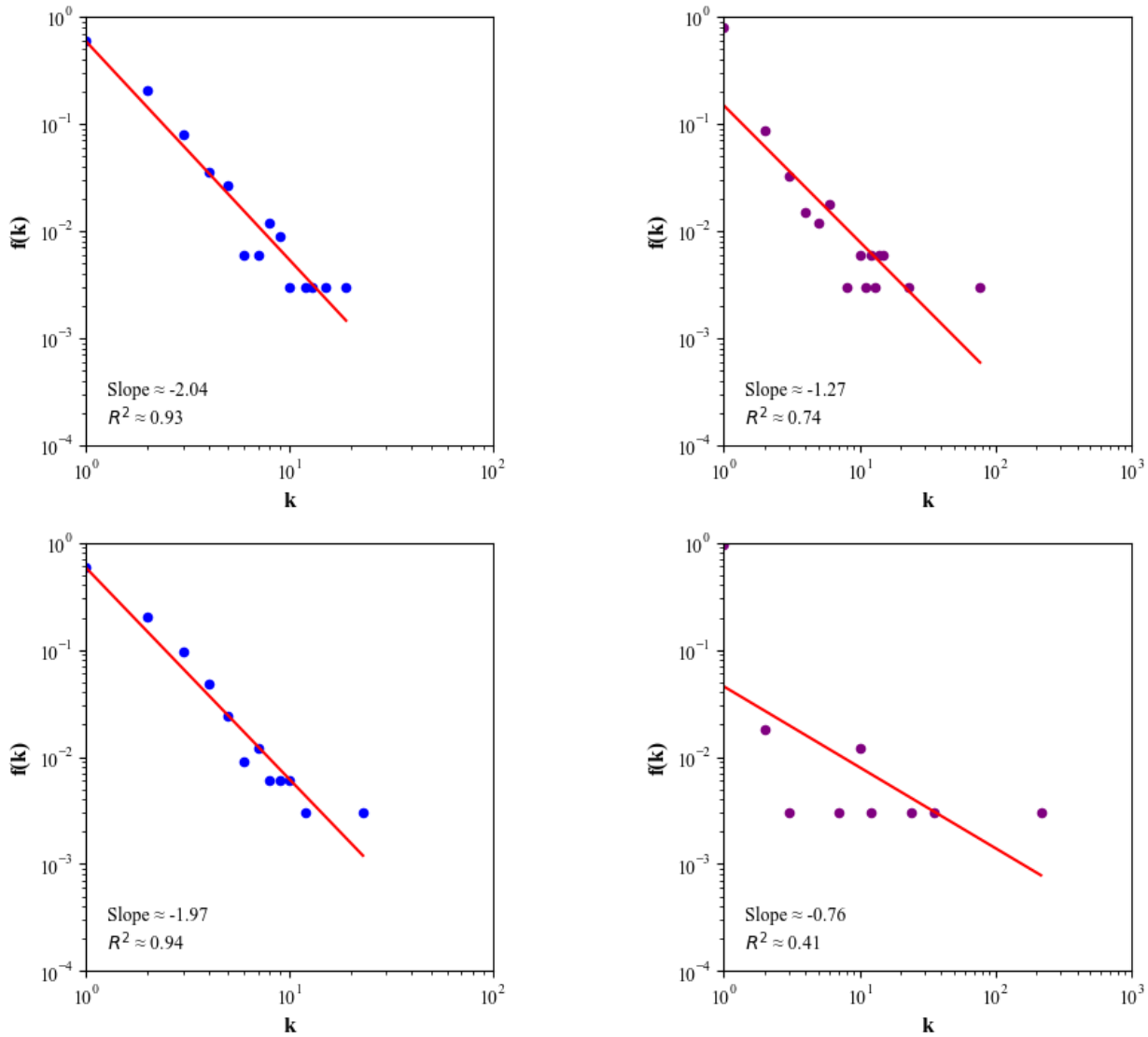

FIG. 3. Comparison of degree distributions during normal and crisis periods. The left panel shows the log-log degree distribution $f(k)$ of the MSTs for the empirical S&P 500 market, while the right panel shows the corresponding the one-factor universe.

In the real market, the degree distribution of the MST is well approximated by a power

law $f(k) \sim k^{-\alpha}$, with exponent $\alpha \approx 2.0 \pm 0.1$ during normal periods and $\alpha \approx 1.8 \pm 0.1$ during crisis periods. The flatter exponent during crises reflects the emergence of a more dominant hub as the tree contracts and stocks move together. The one-factor model produces a qualitatively different distribution. Since all synthetic returns share a common market factor, the resultant MST has few high-degree nodes while most nodes have degree one.

D. Edge Survival Ratio

Figure 4 shows the single-step survival ratio $\sigma(t)$ for the empirical and one-factor MSTs over the full sample period. In both cases, a large majority of edges persist from one window to the next, confirming that the MST topology is broadly stable through time. However, the two universes differ markedly in their baseline level and their response to market events.

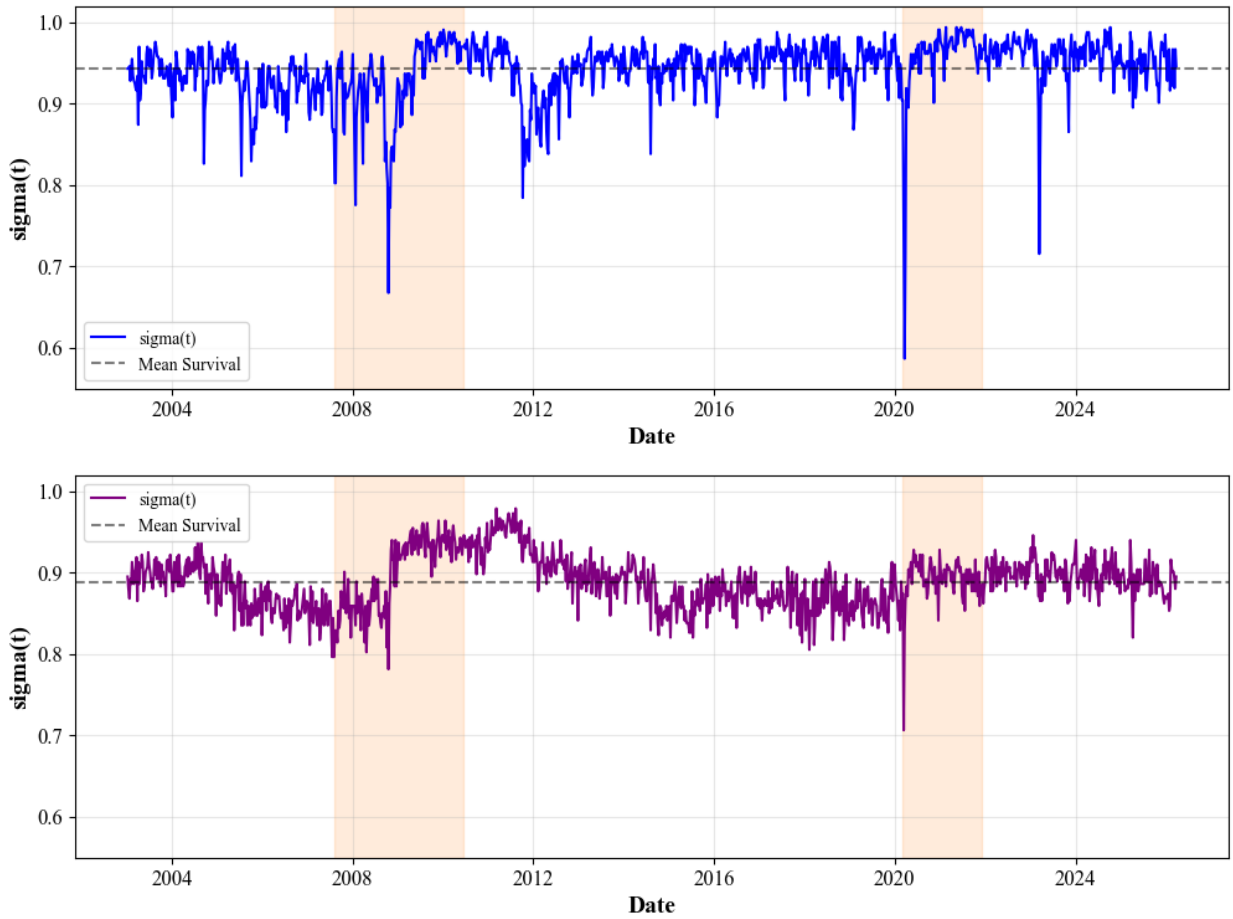

FIG. 4. Comparison of single step survival ratio $\sigma(t)$ as a function of time. The top figure shows the survival ratio for the empirical S&P 500 MSTs, while the bottom figure shows the corresponding one-factor model counterparts. The shaded orange regions correspond to 2 periods of market crises, the 2008 global financial crisis and COVID-19 pandemic of 2020.

The empirical MST maintains a high mean of approximately 0.94, with sharp localised drops at the 2008 global financial crisis and the COVID-19 shock of 2020, where $\sigma(t)$ falls to approximately 0.67 and 0.59 respectively. This reflects rapid topological reorganisation as market-wide correlations rise and the tree restructures around a dominant hub. Apart from these episodes, the survival ratio remains consistently above its mean, confirming that

the sectoral structure of the empirical MST is robust and changes slowly under normal conditions. The one-factor MST exhibits a lower mean of approximately 0.89 and considerably greater variance throughout the sample. During the 2008 crisis it produces only a brief, modest drop before recovering quickly and rising above its mean through the post-crisis period, while the COVID-19 onset produces a sharp dip after which recovery is again rapid.

E. Mean Occupation Layer

Figure 5 shows the mean occupation layer $l(t, v_c)$ for the empirical and one-factor MSTs over the full sample period.

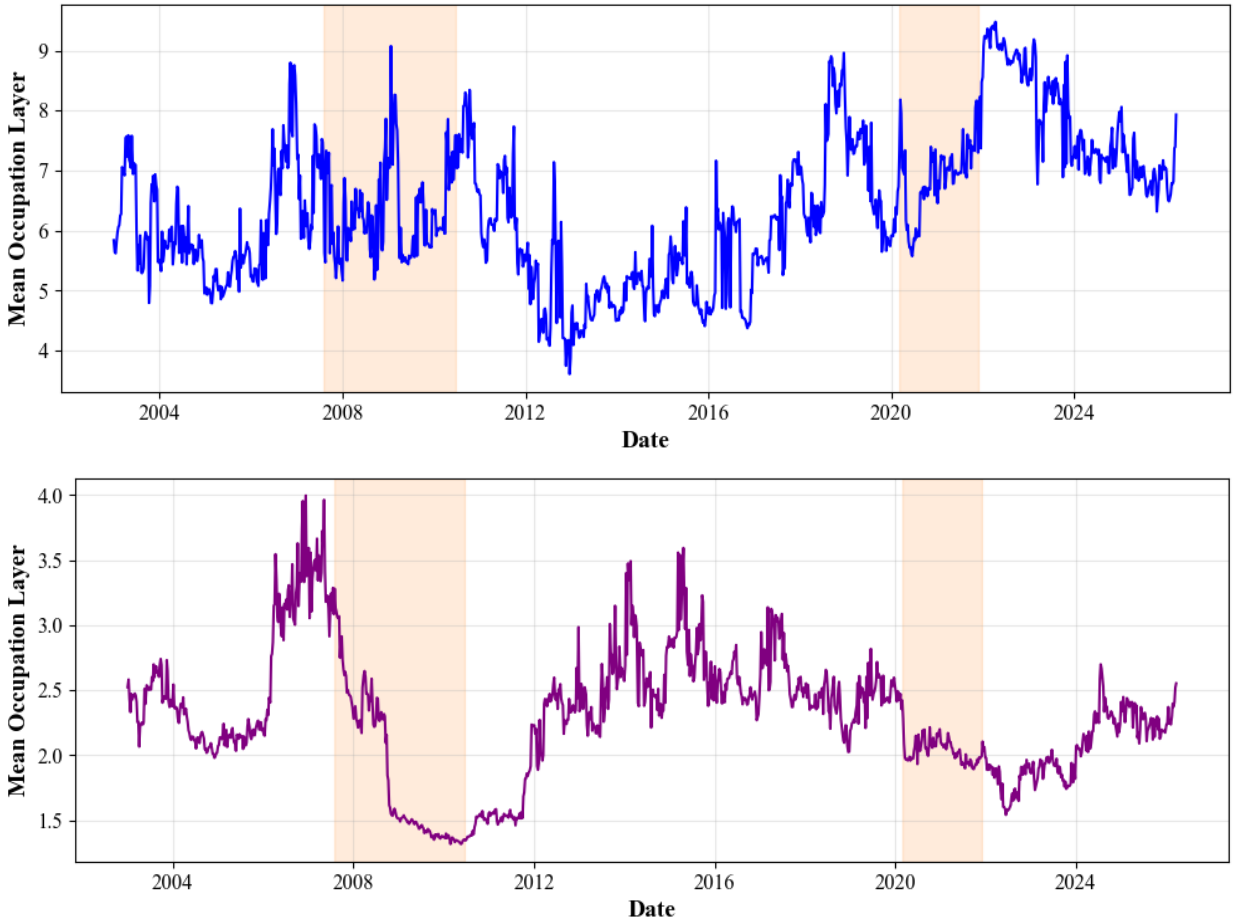

FIG. 5. Comparison of the mean occupation layer $l(t, v_c)$ as a function of time. The top figure shows the MOL for the empirical S&P 500 MST, while the bottom figure shows the one-factor model counterpart. The shaded orange regions correspond to the 2008 global financial crisis and the COVID-19 pandemic of 2020.

The empirical MST maintains a mean occupation layer (MOL) ranging broadly between 4 and 9, reflecting a persistently deep and dispersed tree under normal market conditions. During the 2008 crisis window, the MOL remains flat relative to both the preceding period and the levels it subsequently attains, before contracting further to its sample minimum of approximately 3.7 around 2013. The MOL recovers through 2013 to 2019, dips briefly at the COVID-19 onset in 2020, and rises to its highest values around 2022. The one-factor MST behaves in a starkly different manner: its MOL collapses drastically from approximately 4

to 1.4 during the 2008 crisis and remains depressed through 2012. This is expected, as the contraction reflects the influence of a dominant common factor that pulls all stocks uniformly towards the central hub.

F. Weighted Portfolio Layer

Figure 6 shows the weighted portfolio layer $l_P(t, \theta = 0)$ and the mean occupation layer $l(t, v_c)$ for the empirical and one-factor MSTs over the full sample period. In both cases, the portfolio layer lies above the mean occupation layer for the vast majority of the sample, indicating that the minimum-risk portfolio overweights stocks located further from the central hub than a random stock in the tree.

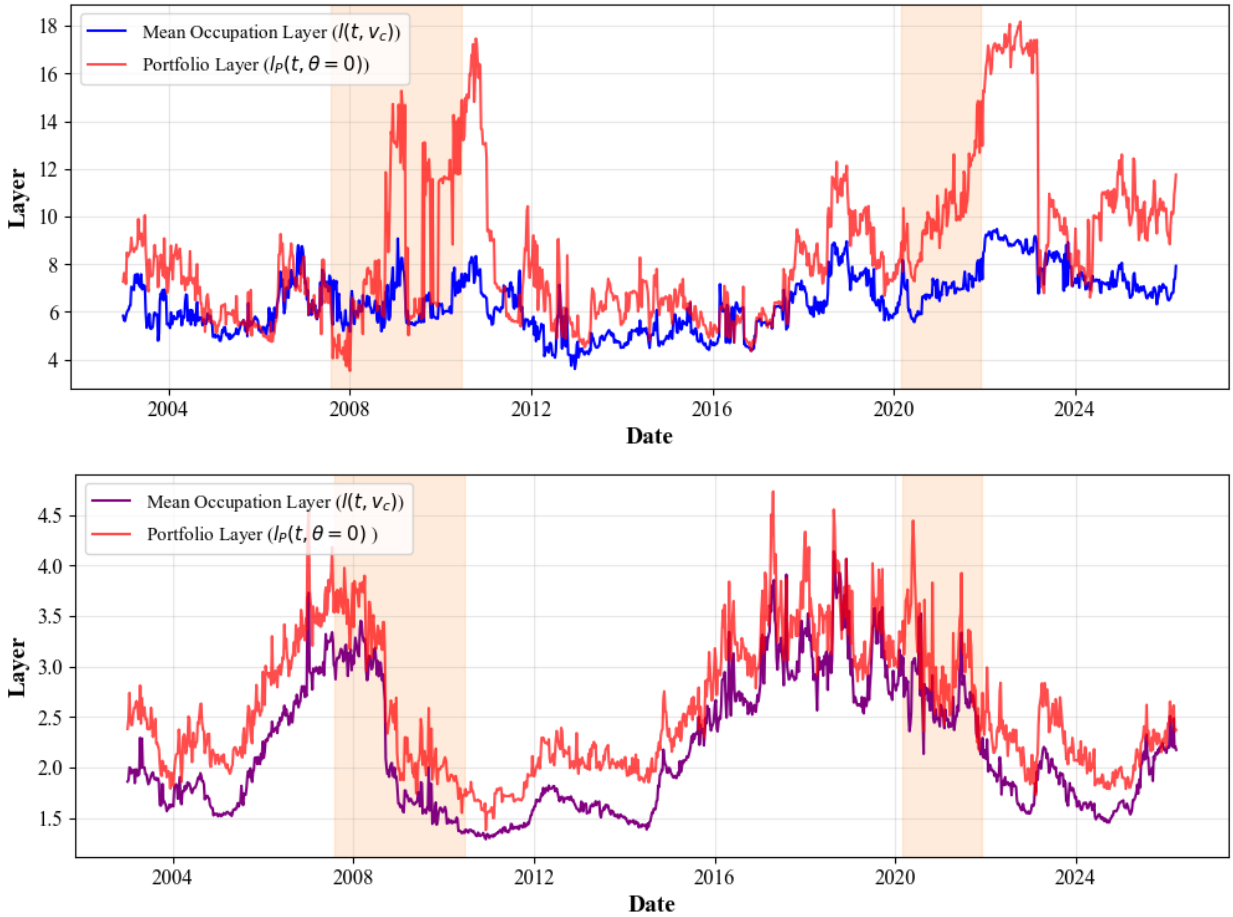

FIG. 6. Comparison of the weighted portfolio layer $l_P(t, \theta = 0)$ and the mean occupation layer $l(t, v_c)$ as functions of time. The top figure shows the weighted portfolio layer for the empirical S&P 500 MST, while the bottom figure shows the one-factor model counterpart. The shaded orange regions correspond to the 2008 global financial crisis and the COVID-19 pandemic of 2020.

In the empirical MST, the separation between the two series is largest during the post-2008 and post-2020 periods, where the portfolio layer rises to considerably higher values while the MOL remains comparatively subdued. This suggests that during periods of market stress, the minimum-risk portfolio is pushed further towards the periphery of the tree as the core becomes more concentrated around the dominant hub. During the troughs of 2012 and 2016, both series converge as the tree contracts and the distance between hub and

peripheral stocks diminishes. In the one-factor MST, the portfolio layer again exceeds the mean occupation layer for most of the sample, but the separation is substantially smaller and the two series follow each other much more closely throughout. This is consistent with the star-like topology of the one-factor universe, where the absence of stable multi-sector sub-trees reduces the structural distinction between central and peripheral stocks.

Overall, the results paint a consistent picture of the S&P 500 correlation network as a hierarchical, multi-sector system whose structure cannot be replicated by the one-factor model. This has direct implications for portfolio construction: the radial position (level) of a stock within the MST carries meaningful information about its role in a minimum-risk portfolio, with peripheral stocks preferred over central, highly connected ones.

IV. Conclusion & Outlook

This project investigated the dynamics of correlation-based financial networks in the S&P 500 from 2000 to 2026, using the minimum spanning tree as a parsimonious representation of equity co-movements across 335 continuously traded stocks. A calibrated one-factor CAPM universe was constructed as a topological null hypothesis, enabling a systemic separation of structure driven by the aggregate market factor from genuine multi-sector economic organisation.

The emprical MST exhibits persistent hierarchical sectoral clustering, a scale-free degree distribution, a robust topology under normal conditions, and a peripheral bias in the minimum-risk portfolio, none of which are replicated in the one-factor universe. Periods of market stress leave clear signals across all analyses: average correlations rise, the survival ratio drops sharply, the mean occupation layer contracts, and the portfolio layer separation widens, collectively reflecting the homogenisation of returns and the concentration of the tree around a dominant hub. The one-factor model captures the elevation in correlations during crises but consistently fails to reproduce the richness of this hierarchical organisation, producing a star-like topology that remains structurally inert across all market conditions. This divergence confirms that the S&P 500 correlation network encodes economically meaningful multi-sector organisation beyond the aggregate market factor, with direct implications for portfolio construction.

Several directions present themselves for future work. The one-factor CAPM baseline could be extended to multi-factor model incorporating size, value and momentum factors, which would provide a more rigorous null hypothesis and isolate the fraction of network structure attributable to well-known risk premia. The portfolio analysis could be entended to the full mean-variance frontier by varying $\theta \in [0, 1]$, testing whether the peripheral bias of minimum-risk stocks weakens monotonically as the target return increases, as suggested by Onnela at al. [4]. The entropy based measures of Kukreti [12] and Chakraborti [10] could be integrated alongside the topological measures studied here to better define the market state and provide early warning signals of market stress. Finally, the framework could be applied to other equity markets or asset classes to assess the universality of the topological results reported here.

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A. Appendixes

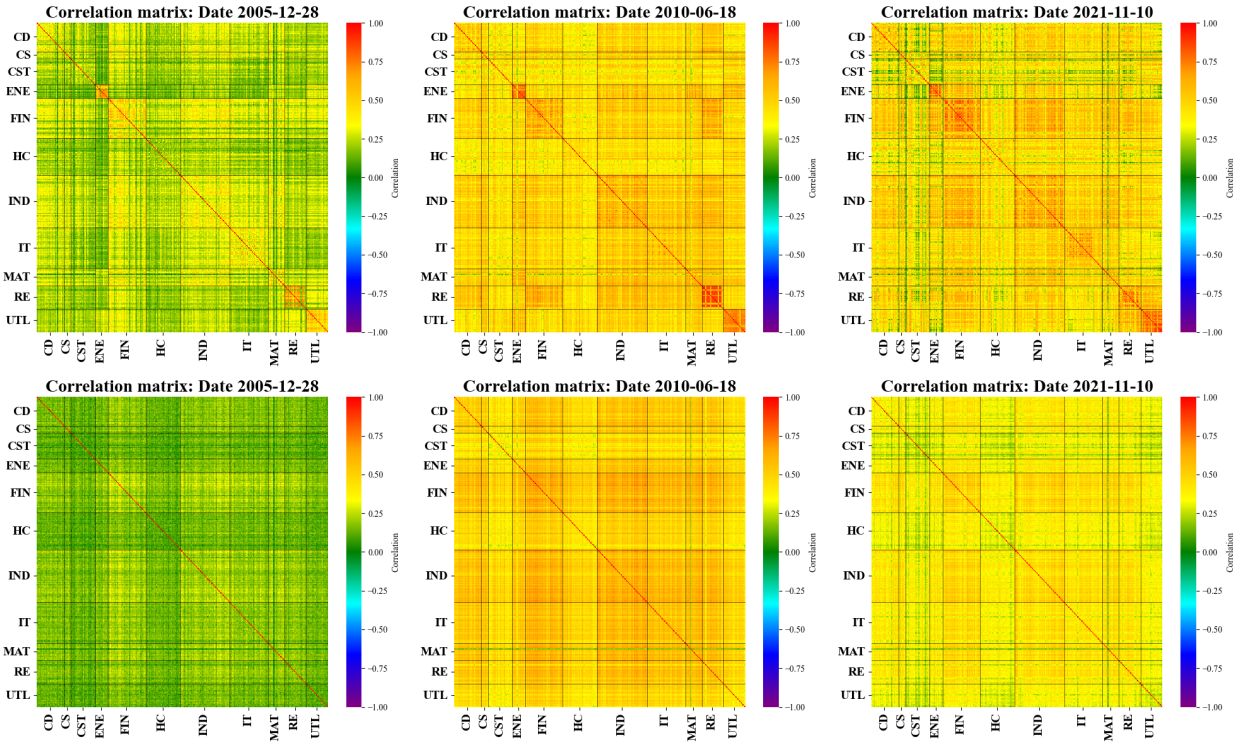

FIG. A1. Comparison of correlation matrices across different periods. The top panel shows the correlation matrices of S&P 500 stocks on 28 December 2005 (normal), 18 June 2010 (crash), and 10 November 2021 (crash). The bottom panel shows the corresponding correlation matrices generated by the one-factor model.

Algorithm A1 Markowitz Portfolio Optimisation

Require: Return matrix $R^t \in \mathbb{R}^{T \times N}$, risk-return parameter $\theta \in [0, 1]$, regularisation parameter $\lambda = 10^{-6}$

Estimate covariance matrix $\Sigma^t \leftarrow \mathrm{Cov}(R^t) + \lambda \mathbf{I}$

Compute minimum return portfolio r_m and maximum return portfolio r_M from the mean-variance frontier

Set target return $r_{P,\theta} \leftarrow (1 - \theta) r_m + \theta r_M$

Solve the quadratic programme:

$$\mathbf{w}^* = \arg\min_{\mathbf{w}} \ \mathbf{w}^\top \Sigma^t \mathbf{w}$$

subject to:

$$\sum_{i=1}^{N} w_i = 1, \qquad w_i \geq 0 \quad \forall\, i, \qquad \mathbf{w}^\top \boldsymbol{\mu}^t = r_{P,\theta}$$

where $\boldsymbol{\mu}^t$ is the vector of mean returns in window t

if solver converges **then**

  $\mathbf{w}^*(\theta) \leftarrow$ solution weights

else

  $\mathbf{w}^*(\theta) \leftarrow$ **nan**

end if

return $\mathbf{w}^*(\theta)$ =0